



Paper ID : 261052

Printed Page: 1

Subject Code:BP813ET

Roll No:

BPHARM

(SEM. VIII) THEORY EXAMINATION 2025-26

PHARMACEUTICAL PRODUCT DEVELOPMENT

TIME: 3 HRS

M.MARKS: 75

Note: 1. Attempt all Sections. If you require any missing data, then choose suitably.

SECTION A

1. Attempt *all* questions in brief.

10 x 2 = 20

a.	Outline the objectives of pharmaceutical product development.
b.	State the significance of stability assessment.
c.	State the applications of non-ionic surfactants in pharmaceutical formulations.
d.	State the functions of two semi-solid excipients.
e.	Mention the types of tablet coating materials with examples.
f.	Define the components of pharmaceutical aerosols.
g.	Name two major optimization software used for pharmaceutical product development.
h.	State the applications of factorial designs.
i.	Name the quality control tests prescribed for pharmaceutical packaging materials.
j.	State the regulatory considerations of packaging materials for pharmaceutical product development.

SECTION B

2. Attempt any *two* parts of the following:

2 x 10 = 20

a.	Elaborate on the role of cyclodextrins in pharmaceutical product development.
b.	Write in brief the types and applications of various pharmaceutical excipients used in the development of NDDS formulations.
c.	Analyze the merits, demerits, and applications of various optimization techniques for pharmaceutical product development with specific examples.

SECTION C

3. Attempt any *five* parts of the following:

7 x 5 = 35

a.	Propose the regulatory aspects related to pharmaceutical formulation development.
b.	Describe the role and applications of polyethylene glycols in pharmaceutical product development.
c.	Explain the roles of suspending and emulsifying agents in pharmaceutical formulations.
d.	Justify the uses of directly compressible pharmaceutical vehicles with examples.
e.	Discuss the method of selection and applications of pharmaceutical excipients with specific industrial applications.
f.	Describe in brief the method of optimization by factorial designs and their applications.
g.	Explain the method of selection of packaging materials for pharmaceutical product development.